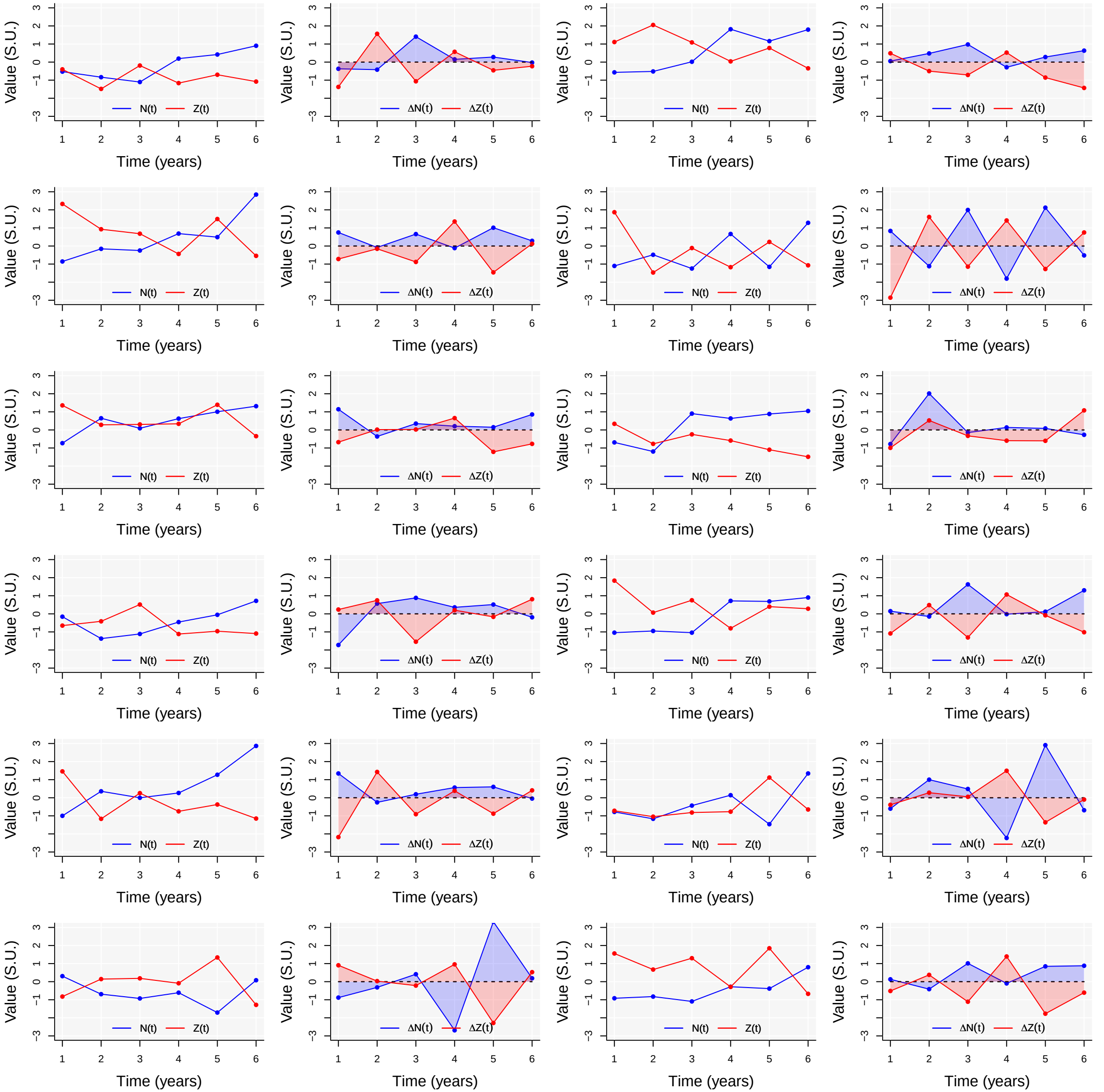
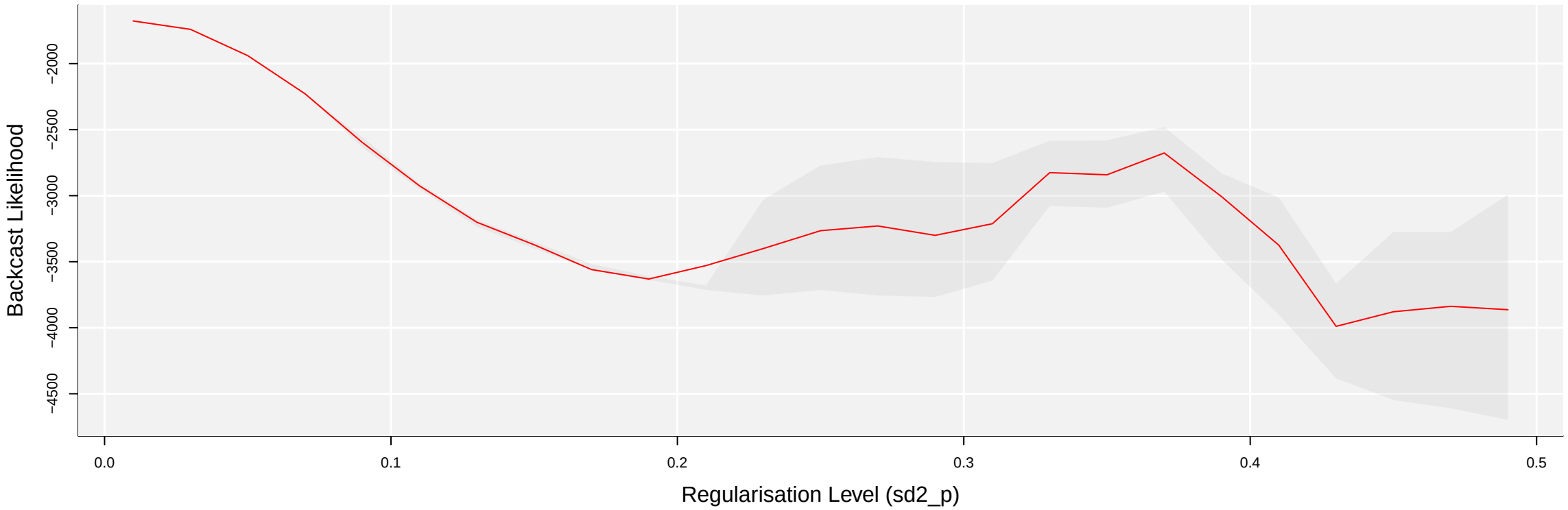
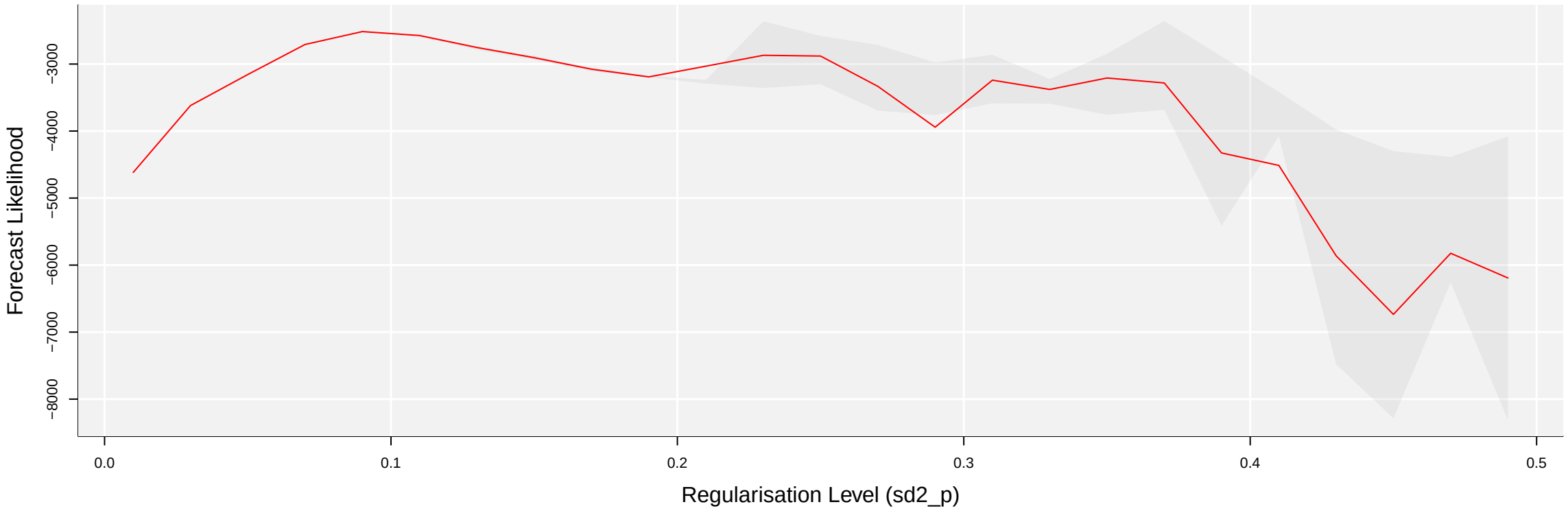
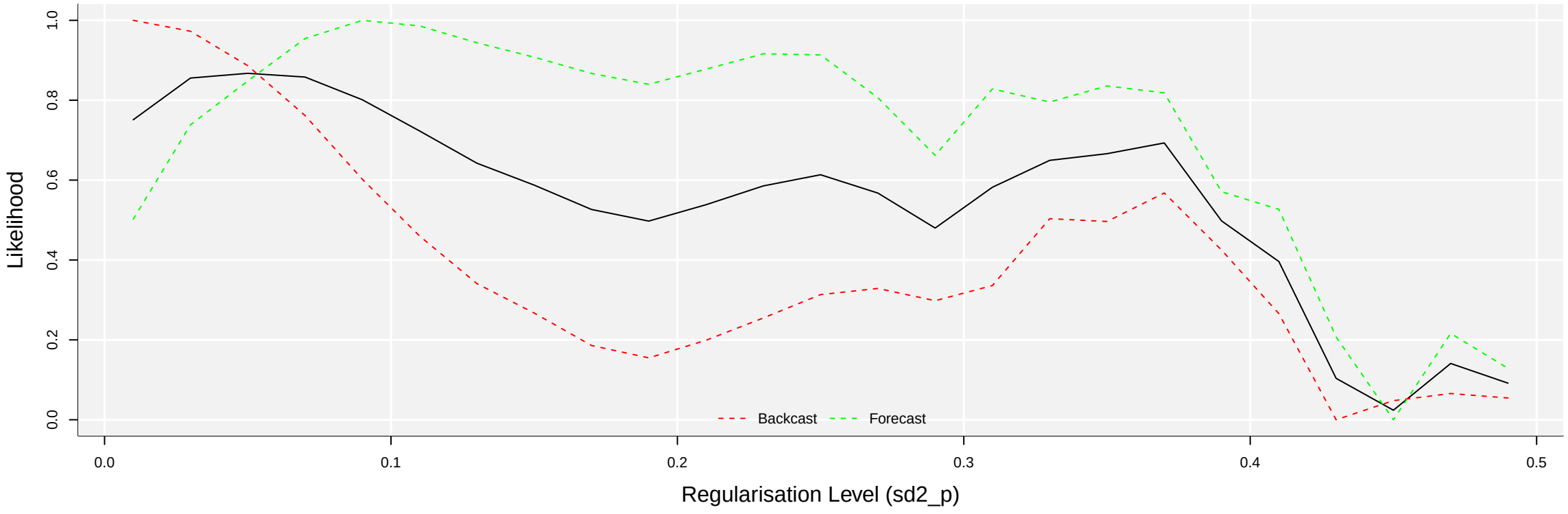
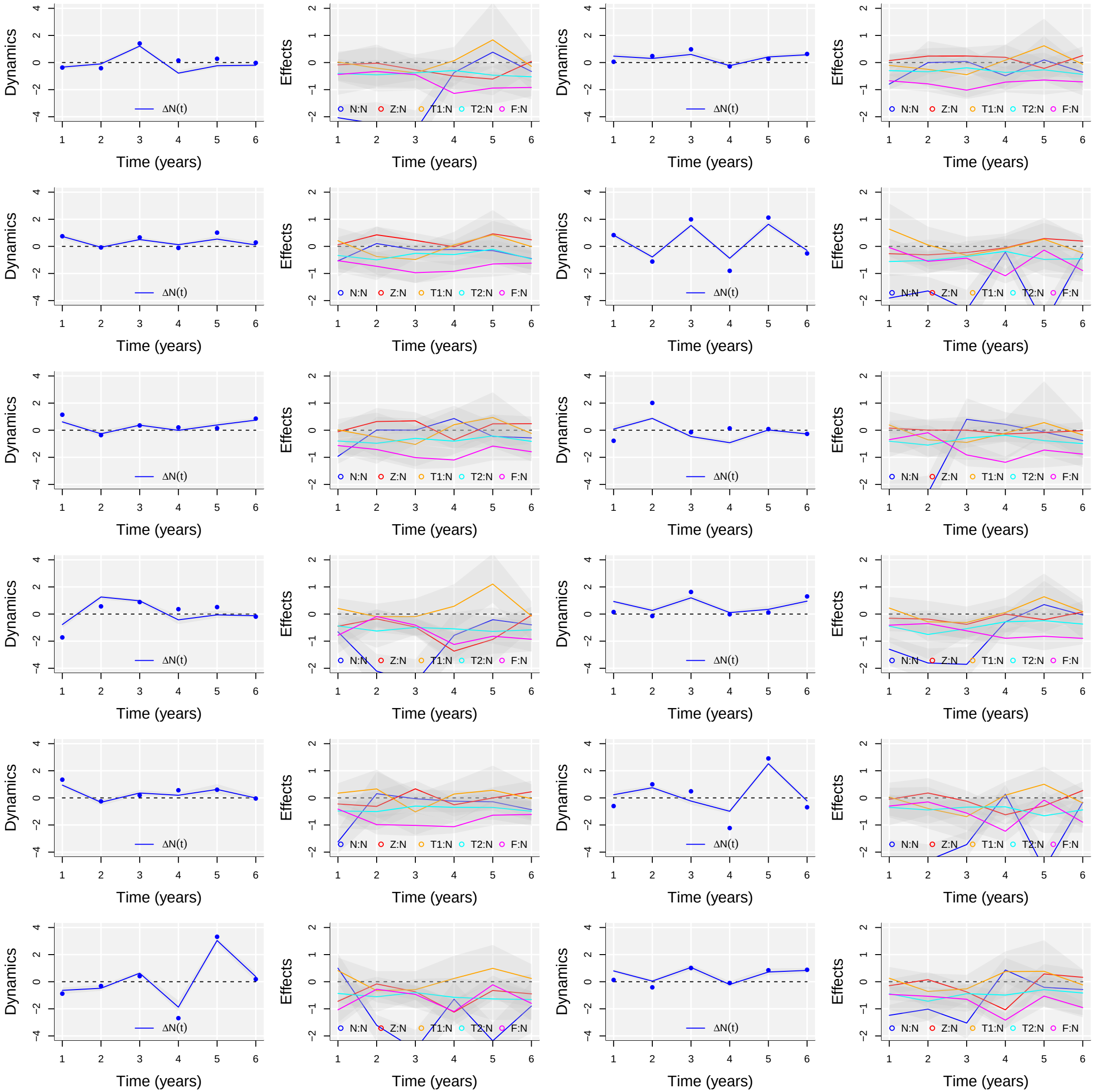
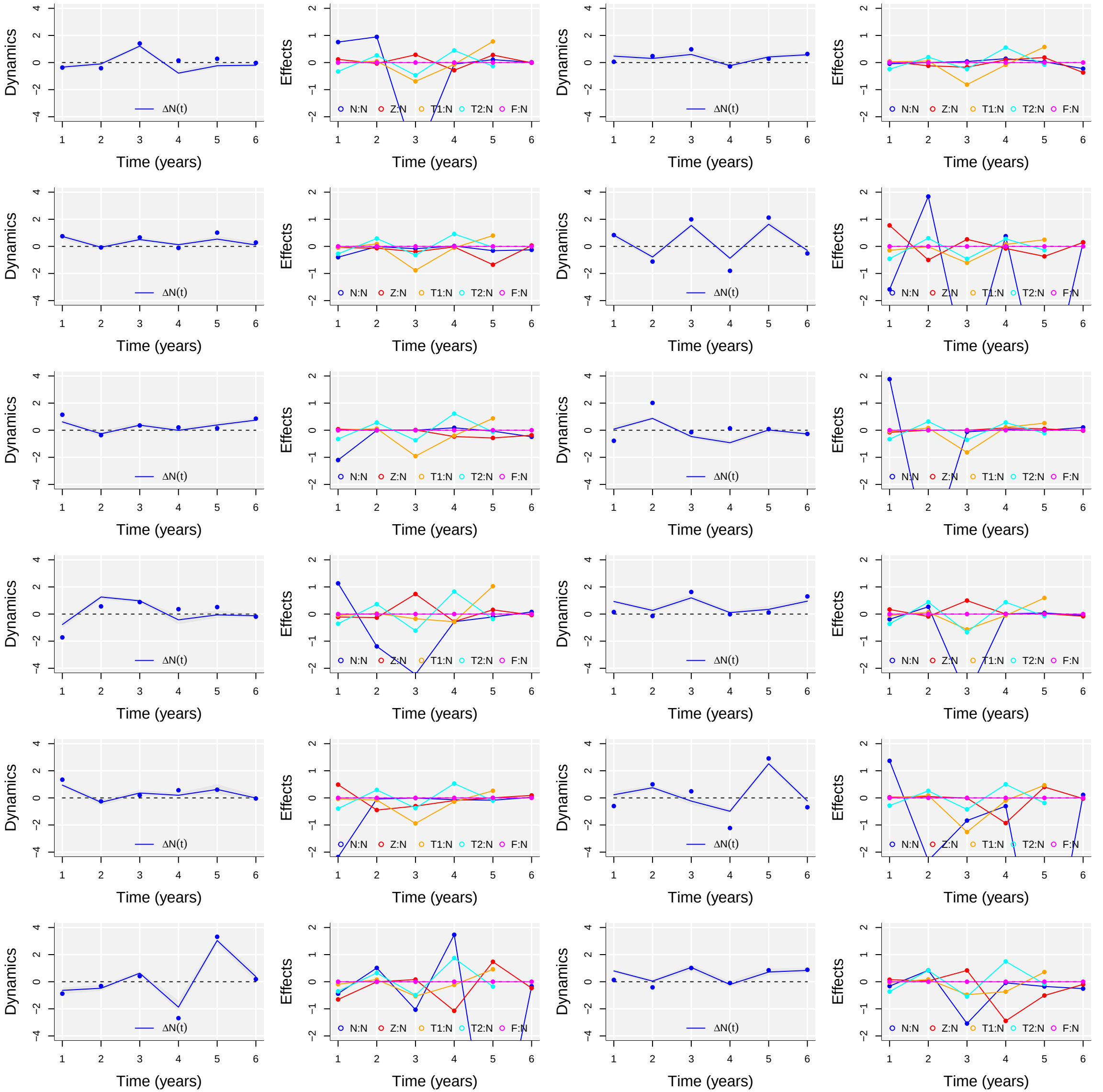

Value (S.U.)
Time (years)
N(t)
Z(t)
ΔN(t)
ΔZ(t)

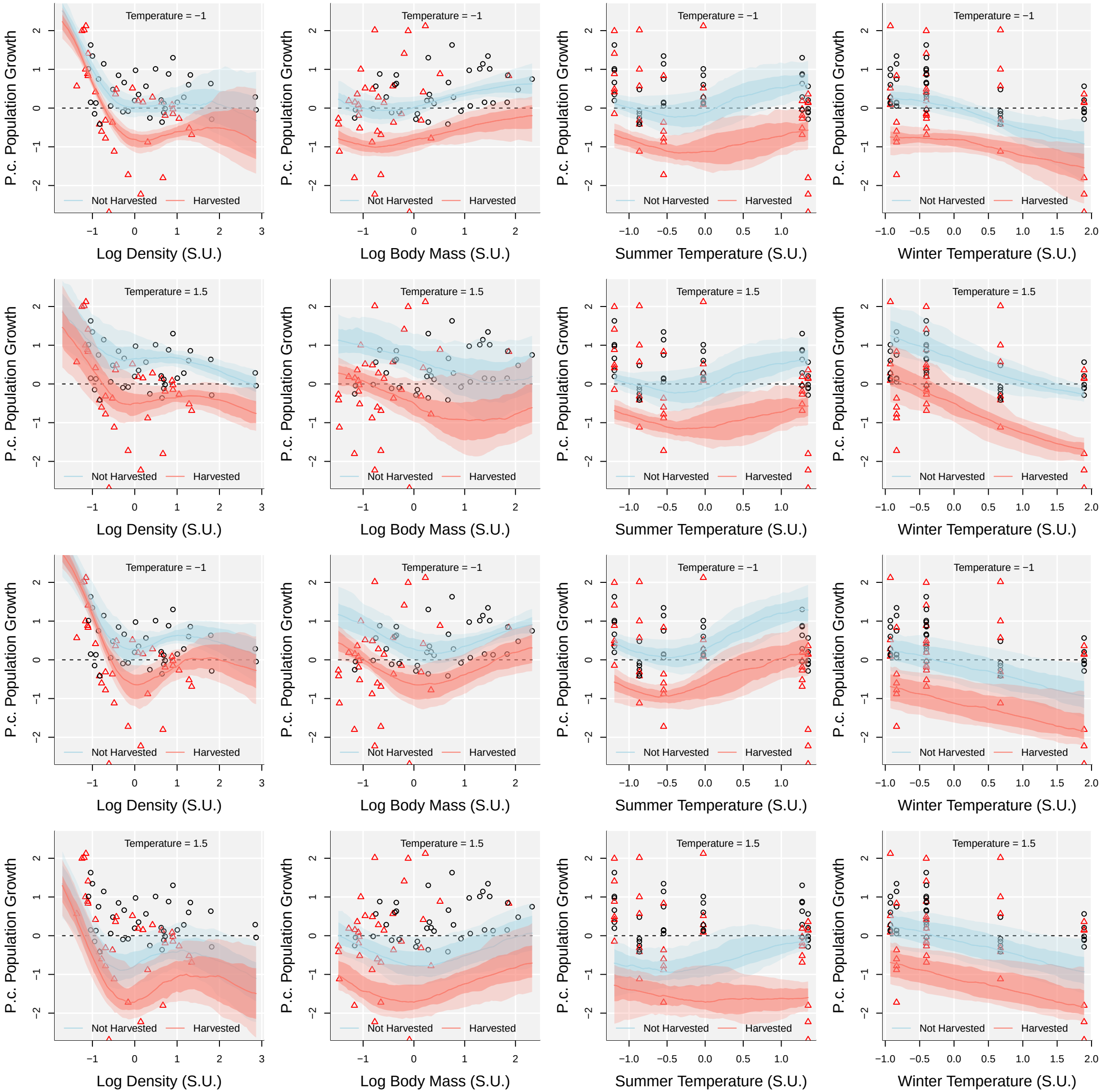

Backcast Likelihood
−2000
−2500
−3000
−3500
−4000
−4500
0.0
0.1
0.2
0.3
0.4
0.5
Regularisation Level (sd2_p)

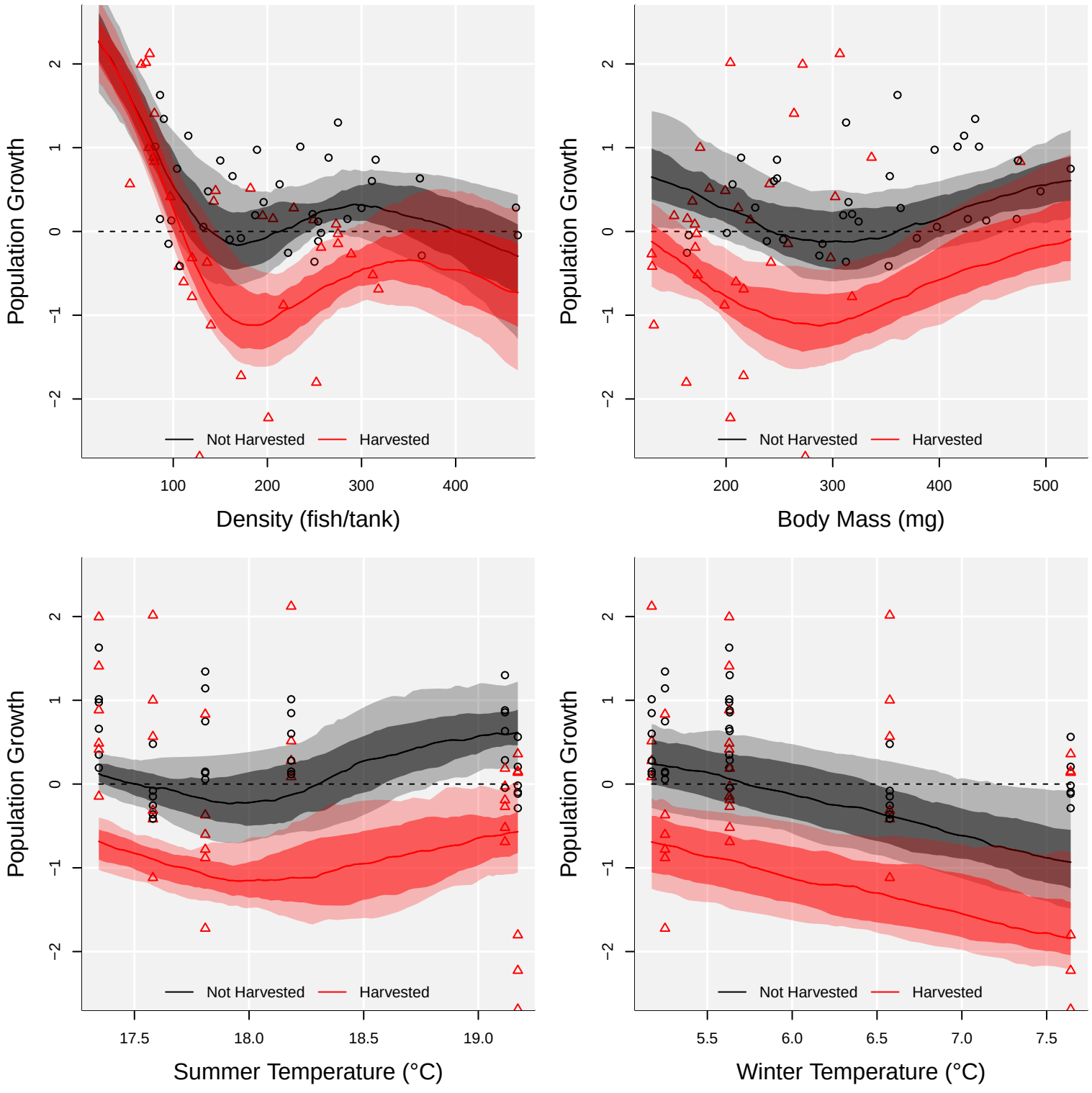

Forecast Likelihood
−3000
−4000
−5000
−6000
−7000
−8000
0.0
0.1
0.2
0.3
0.4
0.5
Regularisation Level (sd2_p)

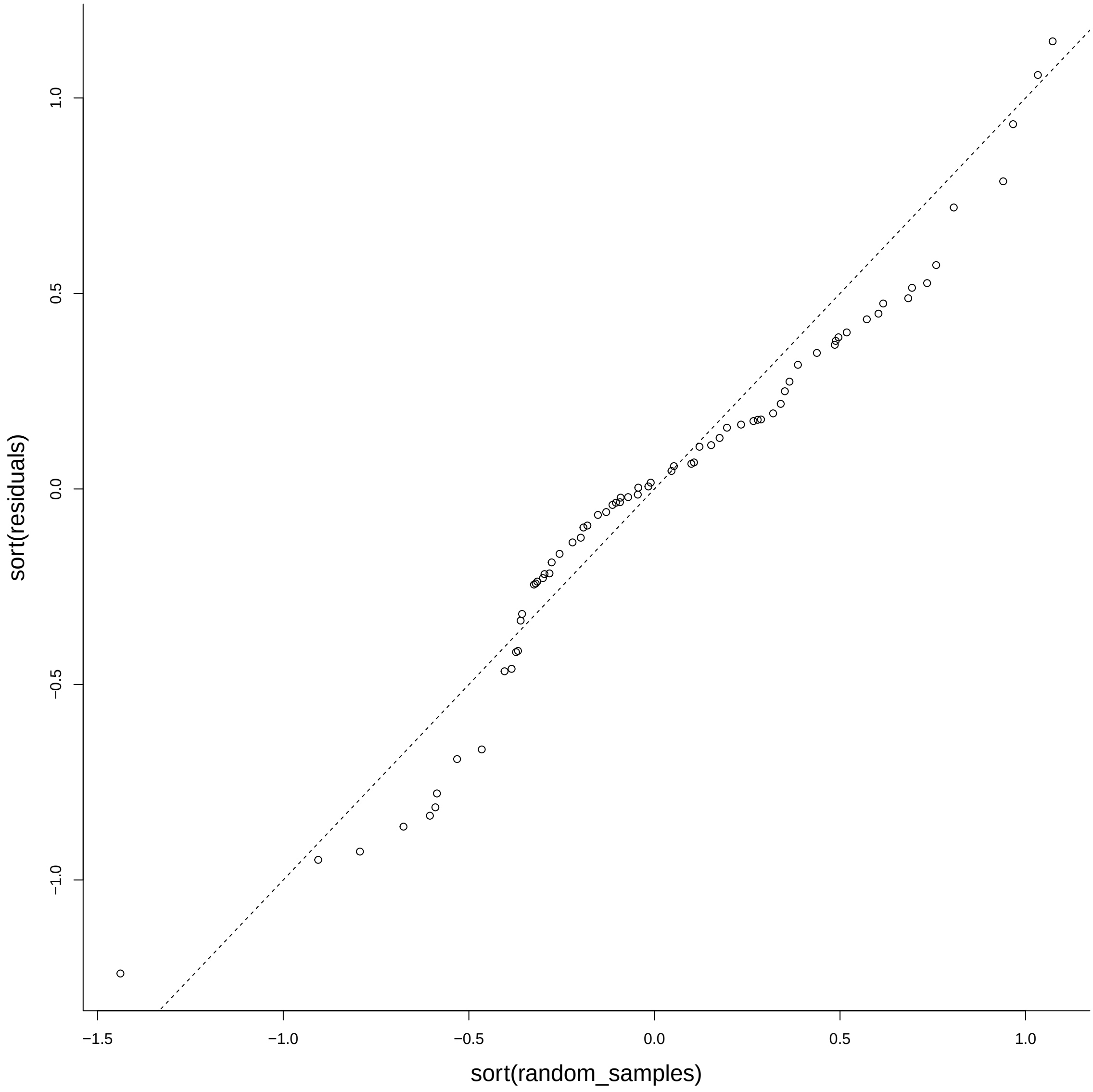

Likelihood
1.0
0.8
0.6
0.4
0.2
0.0
Backcast
Forecast
0.0
0.1
0.2
0.3
0.4
0.5
Regularisation Level (sd2_p)

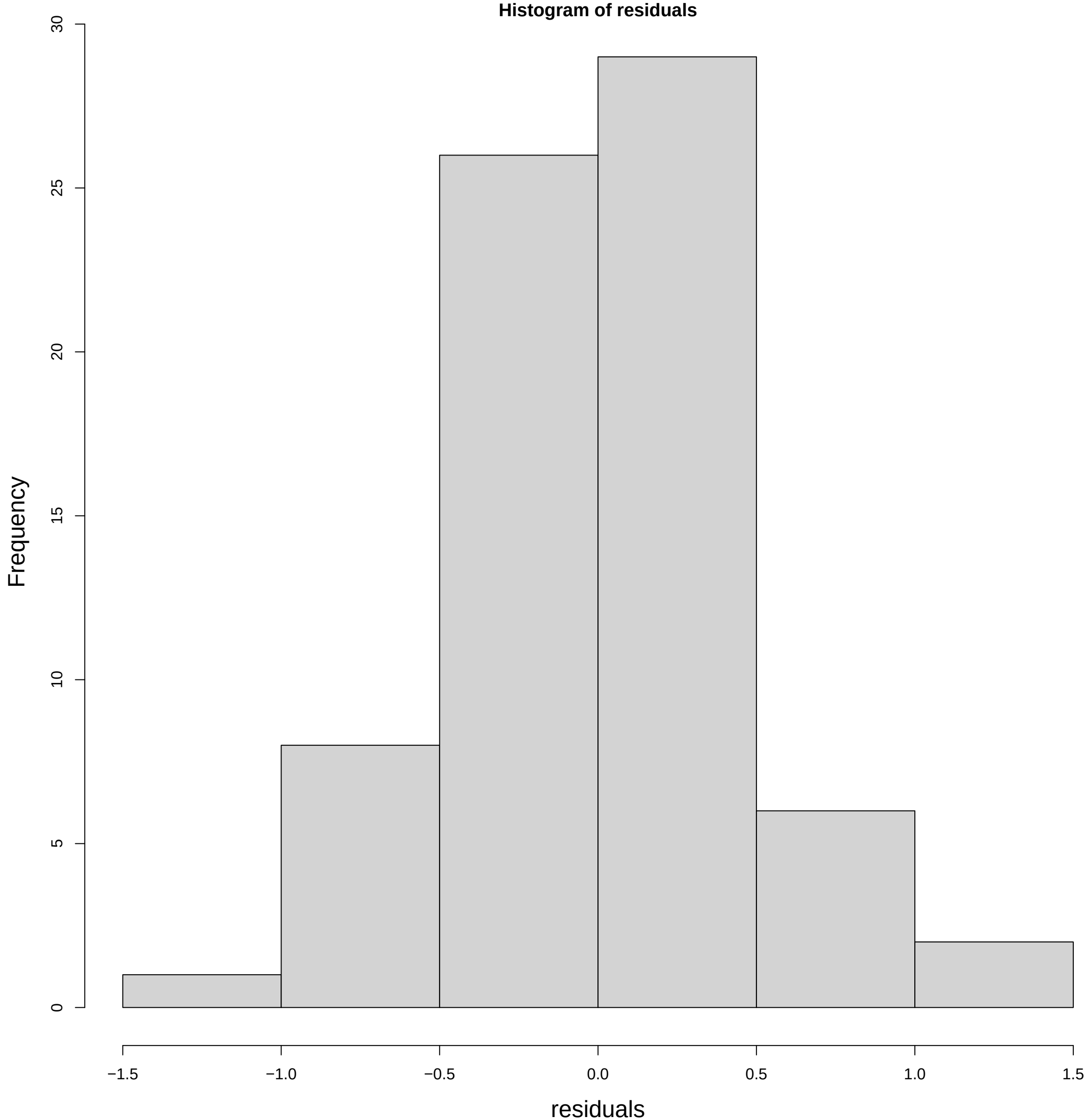

Dynamics
Effects
Time (years)
ΔN(t)
N:N
Z:N
T1:N
T2:N
F:N

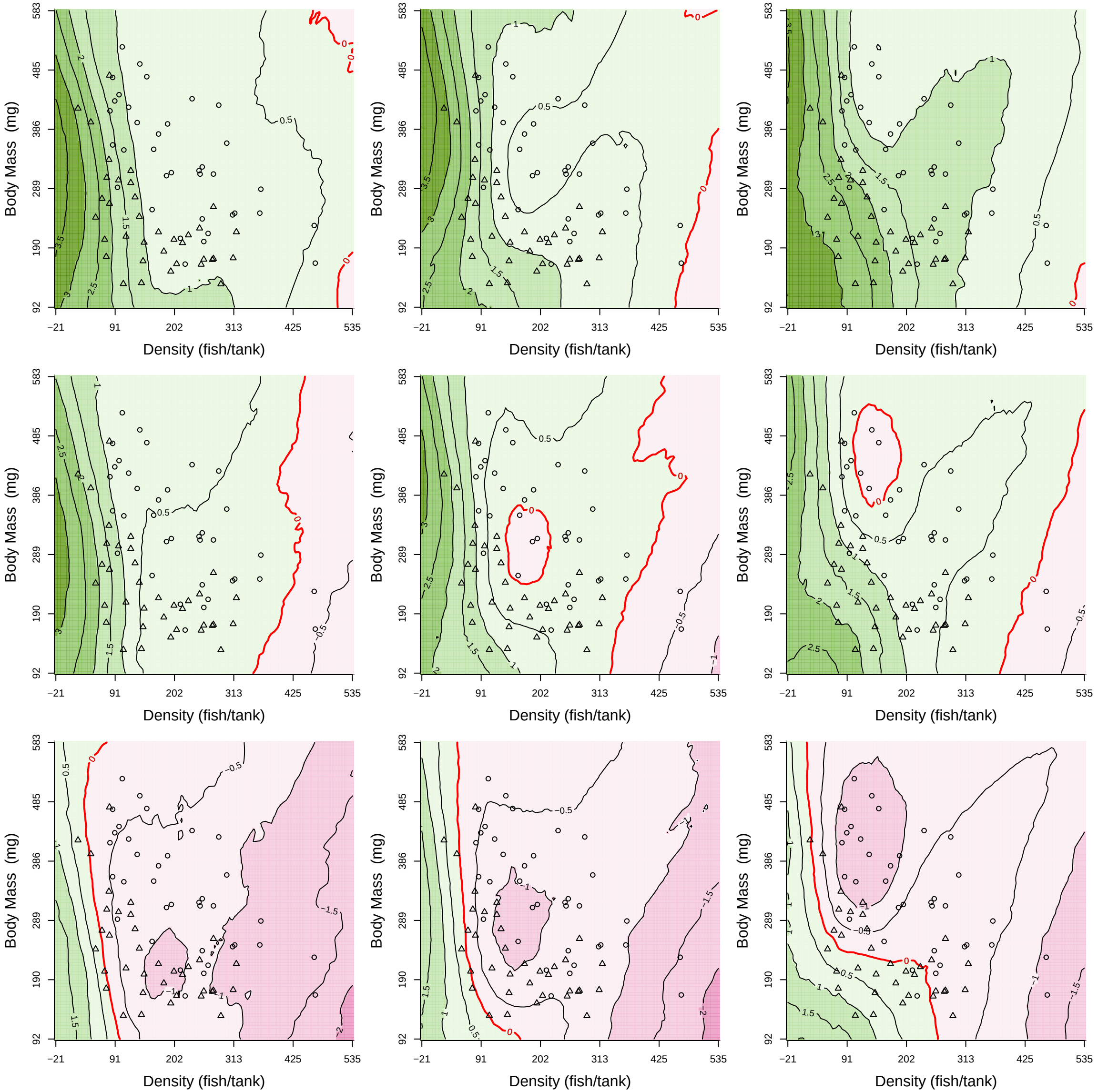

Dynamics
Effects
Time (years)
ΔN(t)
N:N
Z:N
T1:N
T2:N
F:N

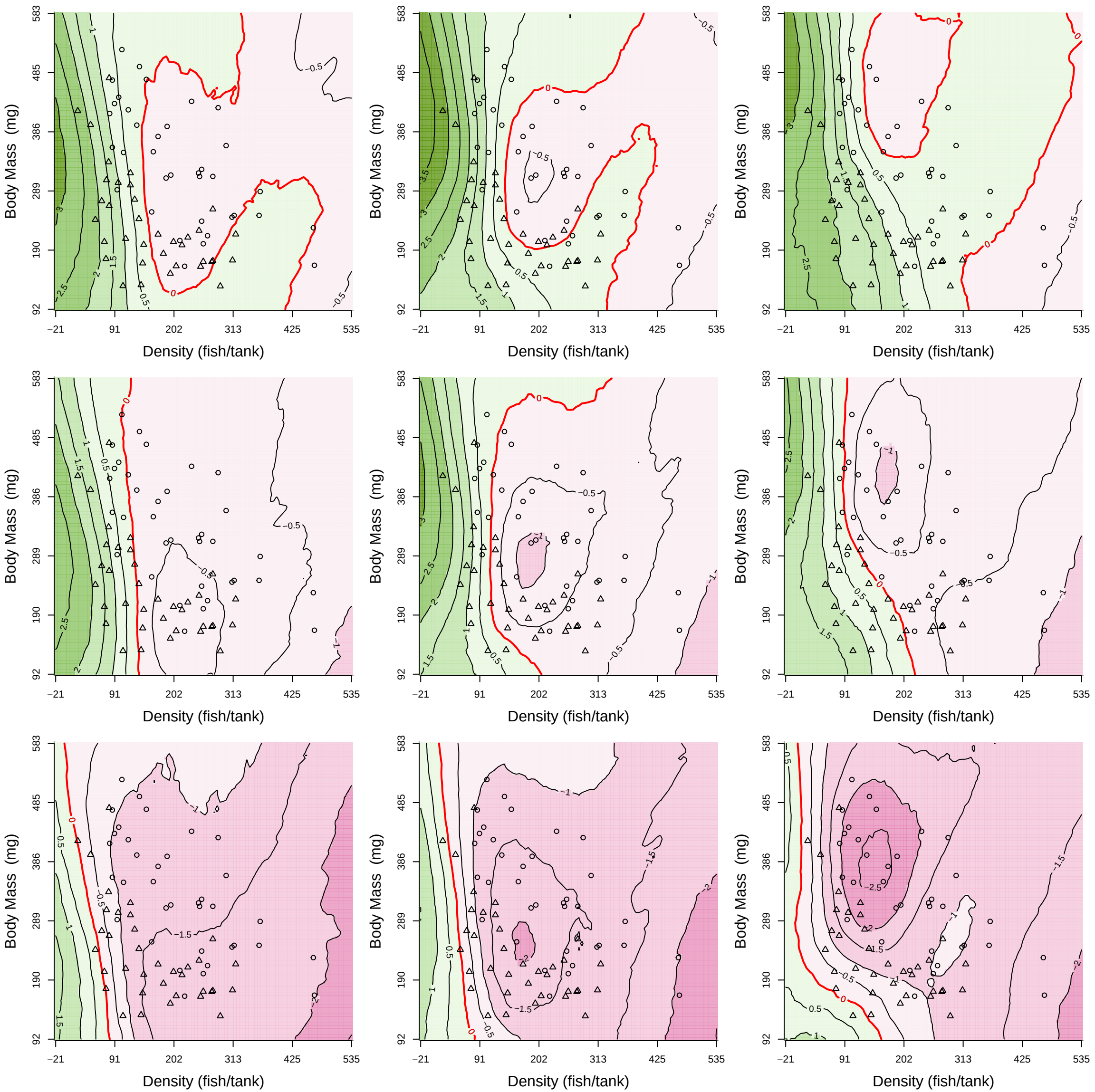

Temperature = −1
Temperature = 1.5
P.c. Population Growth
Not Harvested
Harvested
Log Density (S.U.)
Log Body Mass (S.U.)
Summer Temperature (S.U.)
Winter Temperature (S.U.)

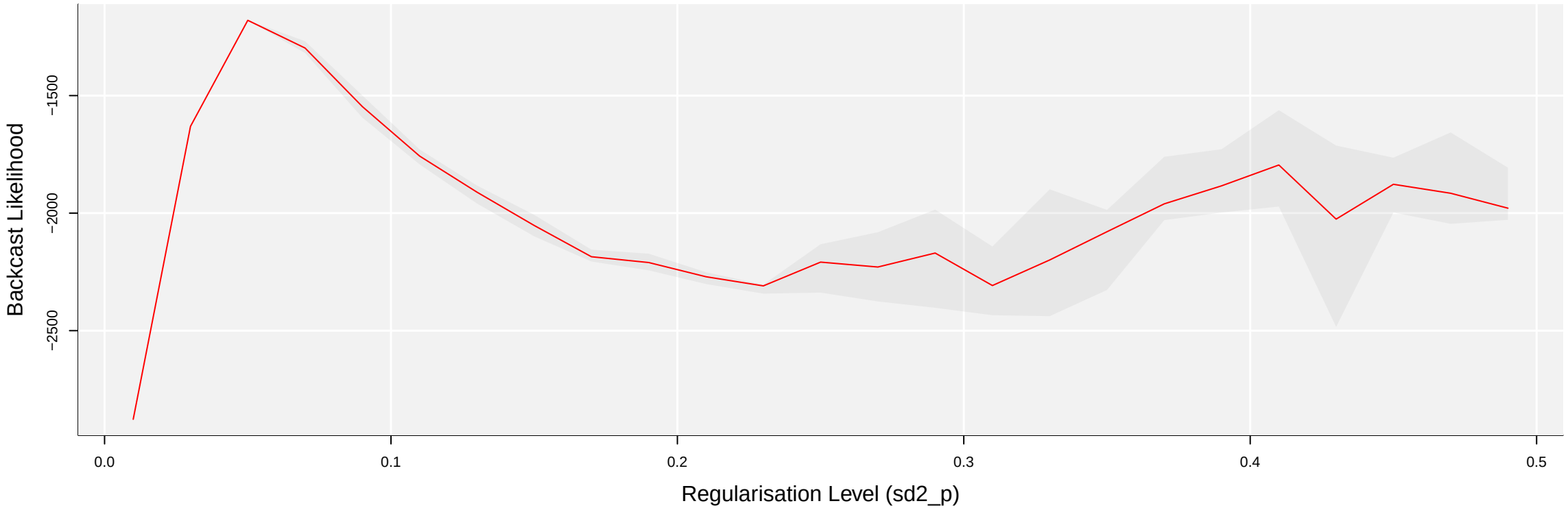

Population Growth
Density (fish/tank)
Not Harvested
Harvested
Population Growth
Body Mass (mg)
Not Harvested
Harvested
Population Growth
Summer Temperature (°C)
Not Harvested
Harvested
Population Growth
Winter Temperature (°C)
Not Harvested
Harvested

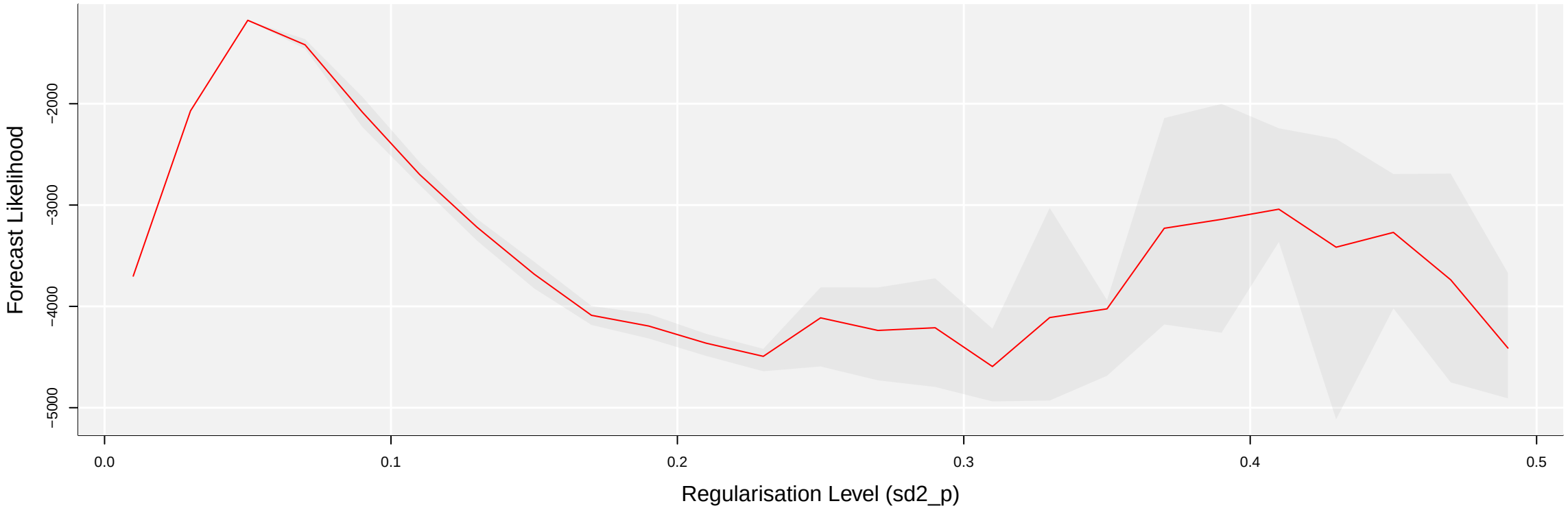

sort(residuals)
1.0
0.5
0.0
−0.5
−1.0
−1.5
−1.0
−0.5
0.0
0.5
1.0
sort(random_samples)

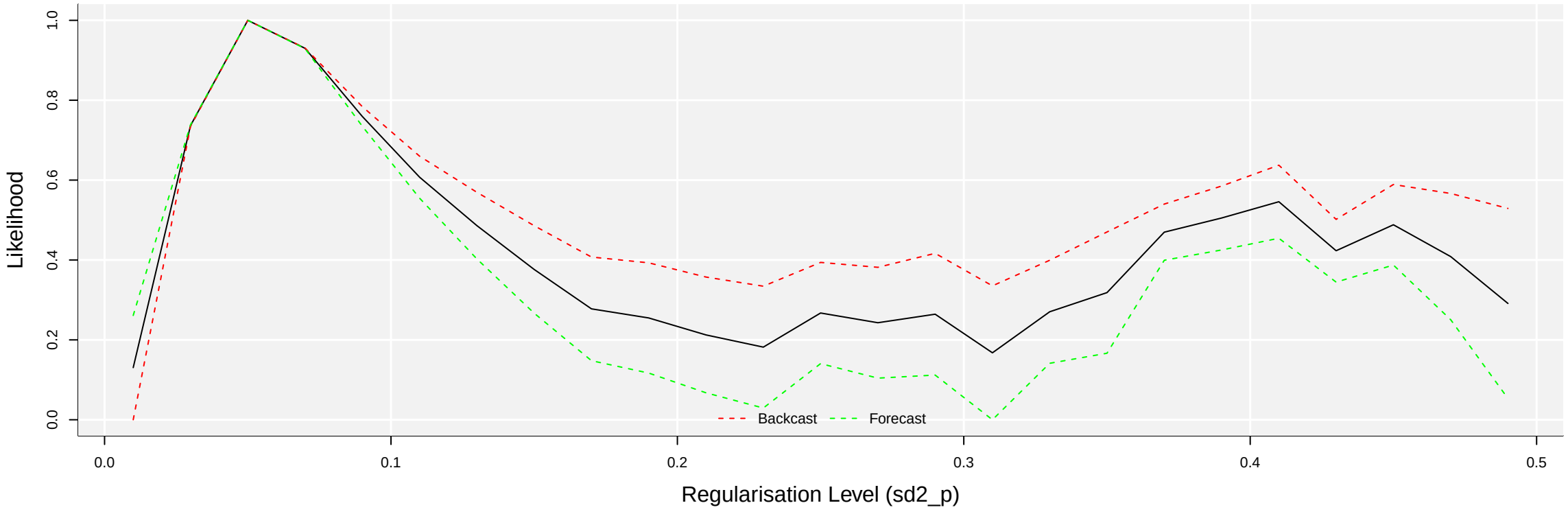

Histogram of residuals
Frequency
0
5
10
15
20
25
30
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
residuals

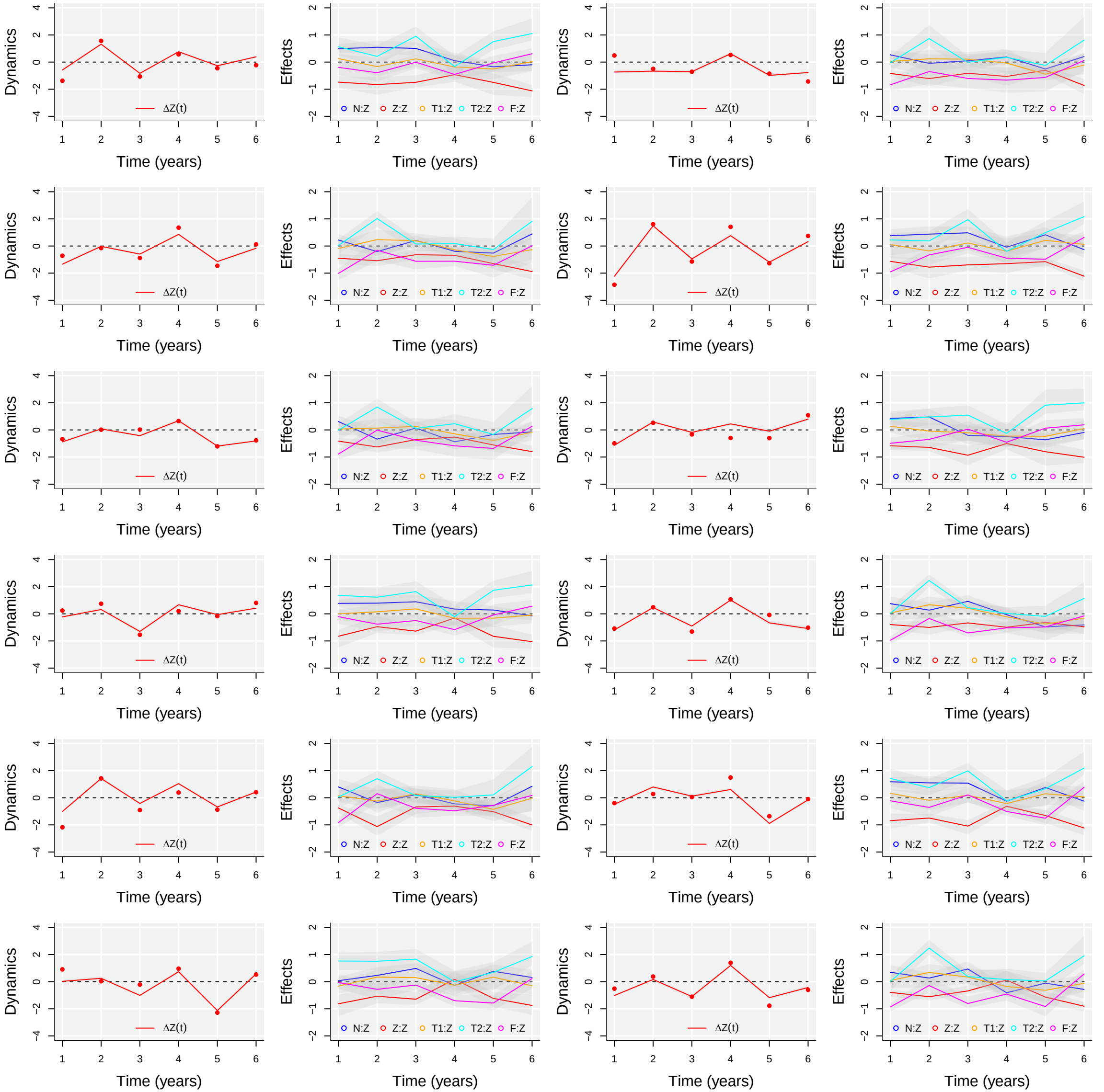

Body Mass (mg)
Density (fish/tank)
583
485
386
289
190
92
−21
91
202
313
425
535

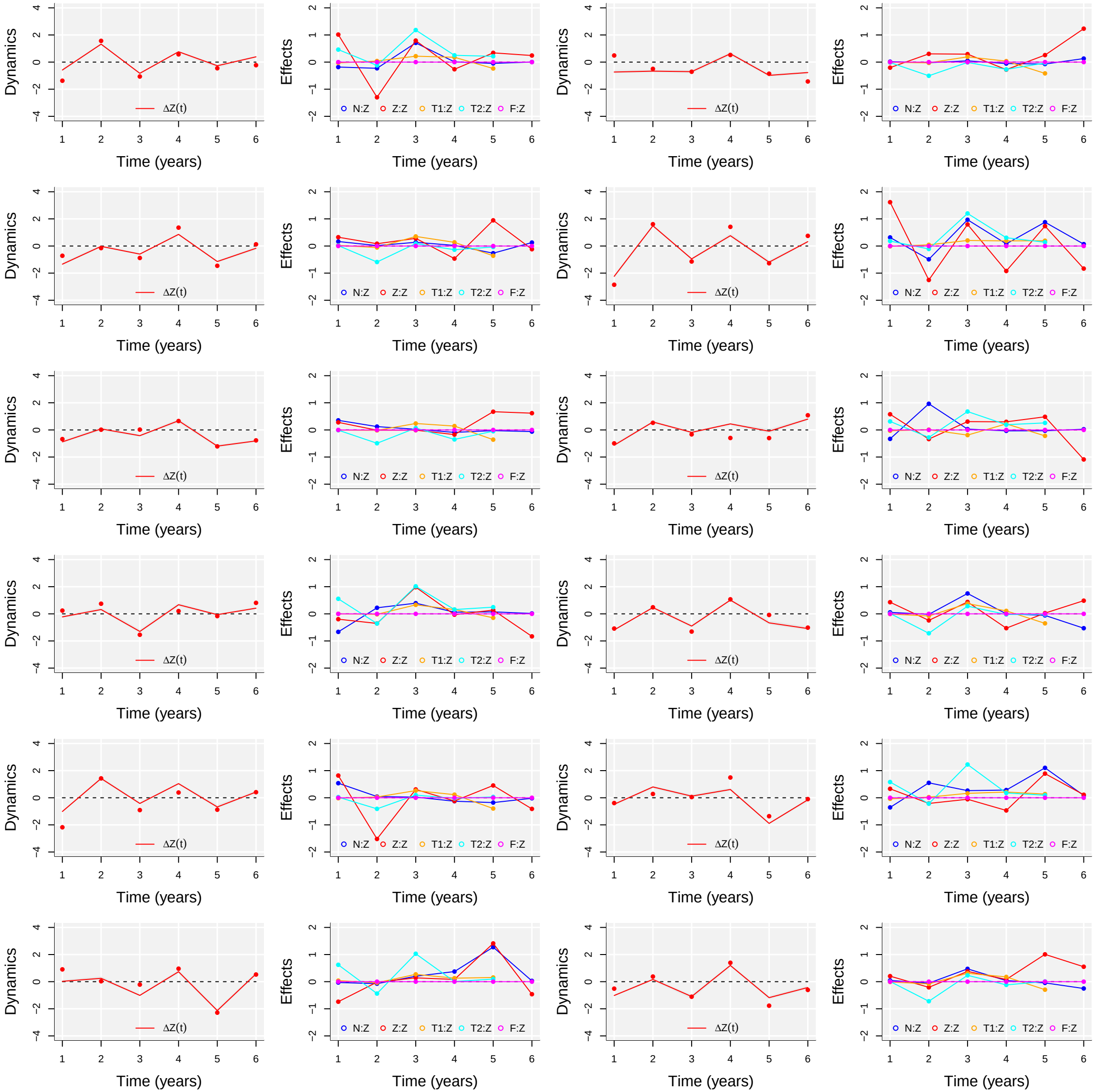

Body Mass (mg)
Density (fish/tank)
583
485
386
289
190
92
−21
91
202
313
425
535

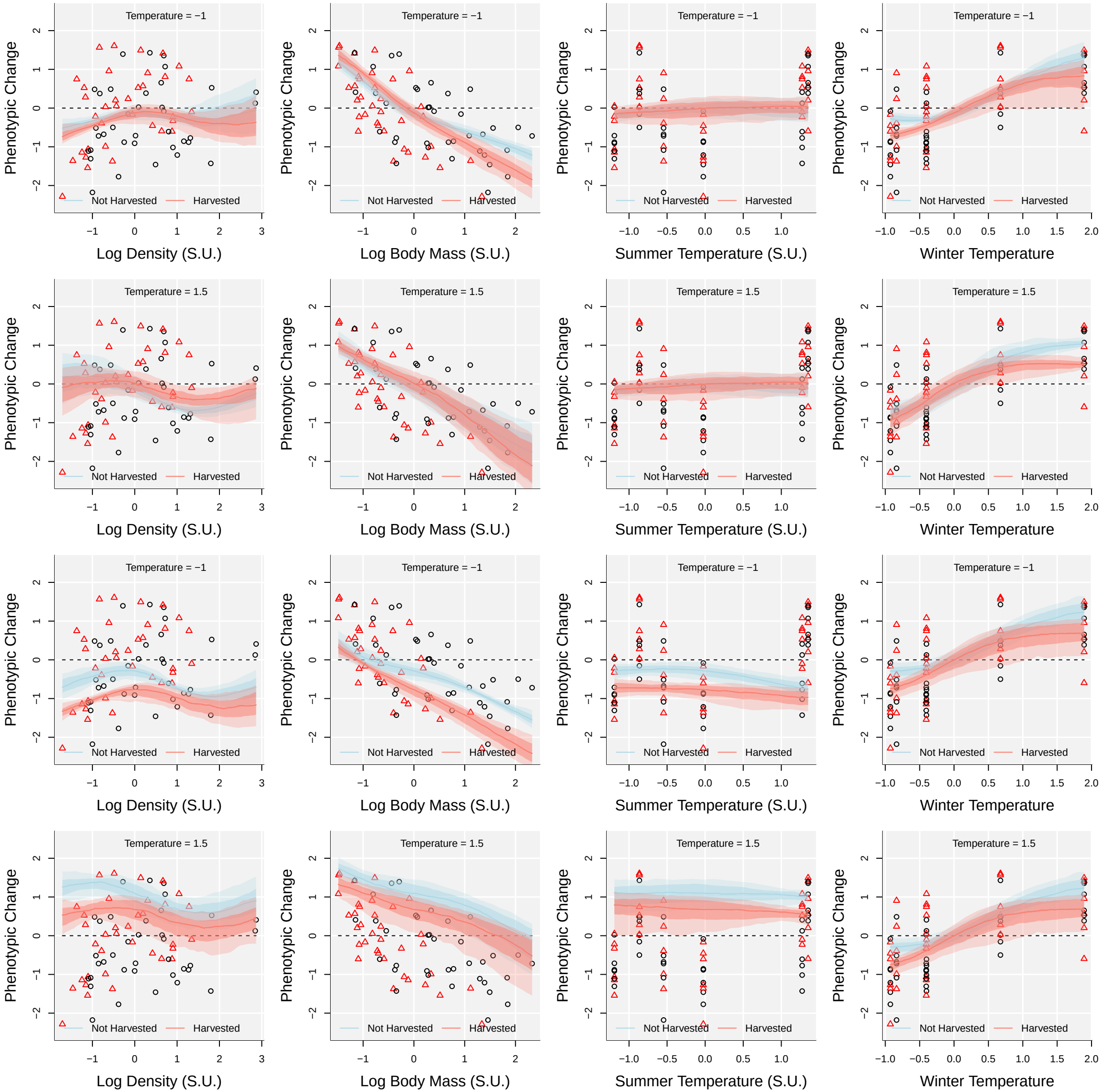

Backcast Likelihood
−1500
−2000
−2500
0.0
0.1
0.2
0.3
0.4
0.5
Regularisation Level (sd2_p)

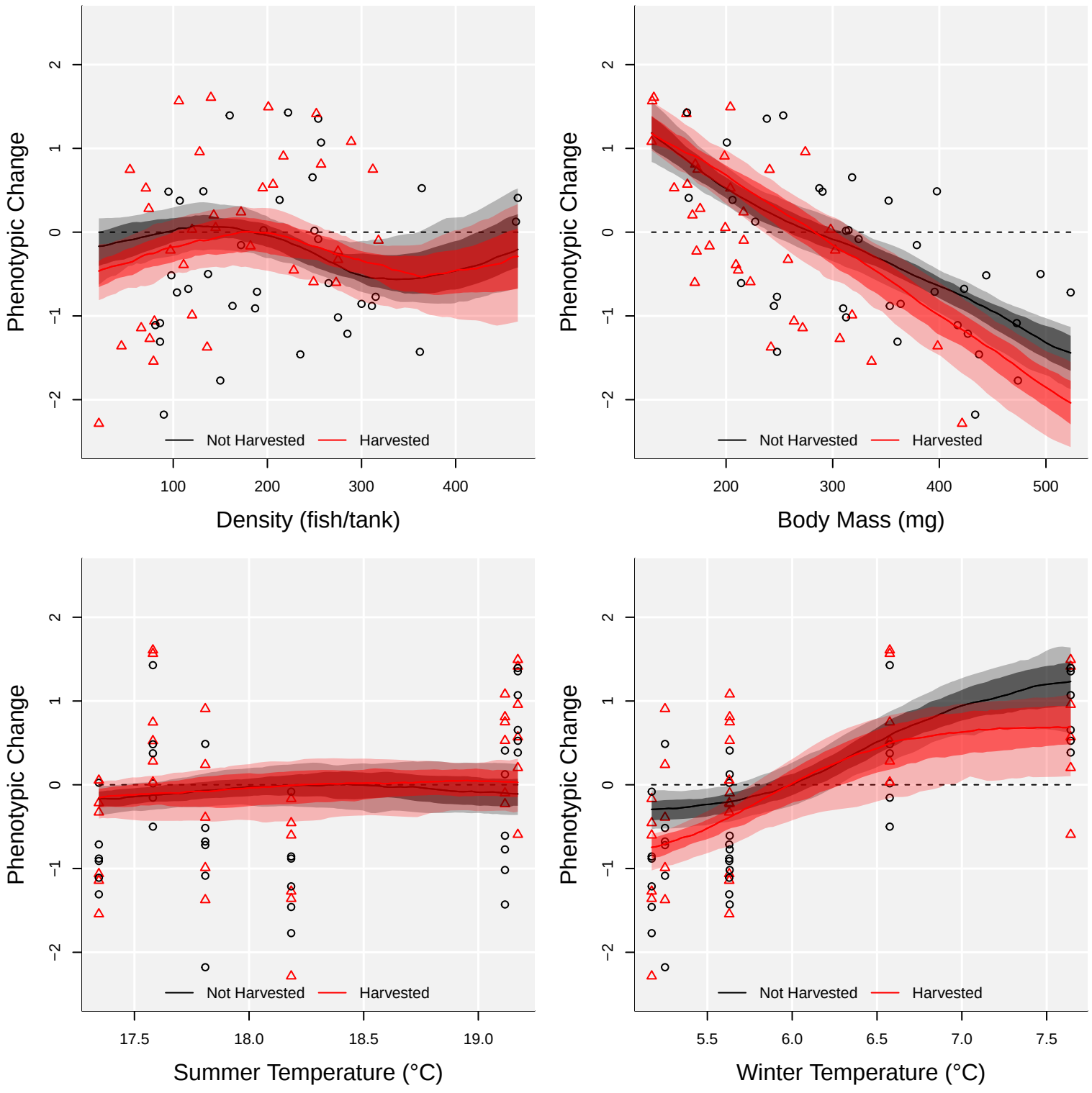

Forecast Likelihood
−2000
−3000
−4000
−5000
0.0
0.1
0.2
0.3
0.4
0.5
Regularisation Level (sd2_p)

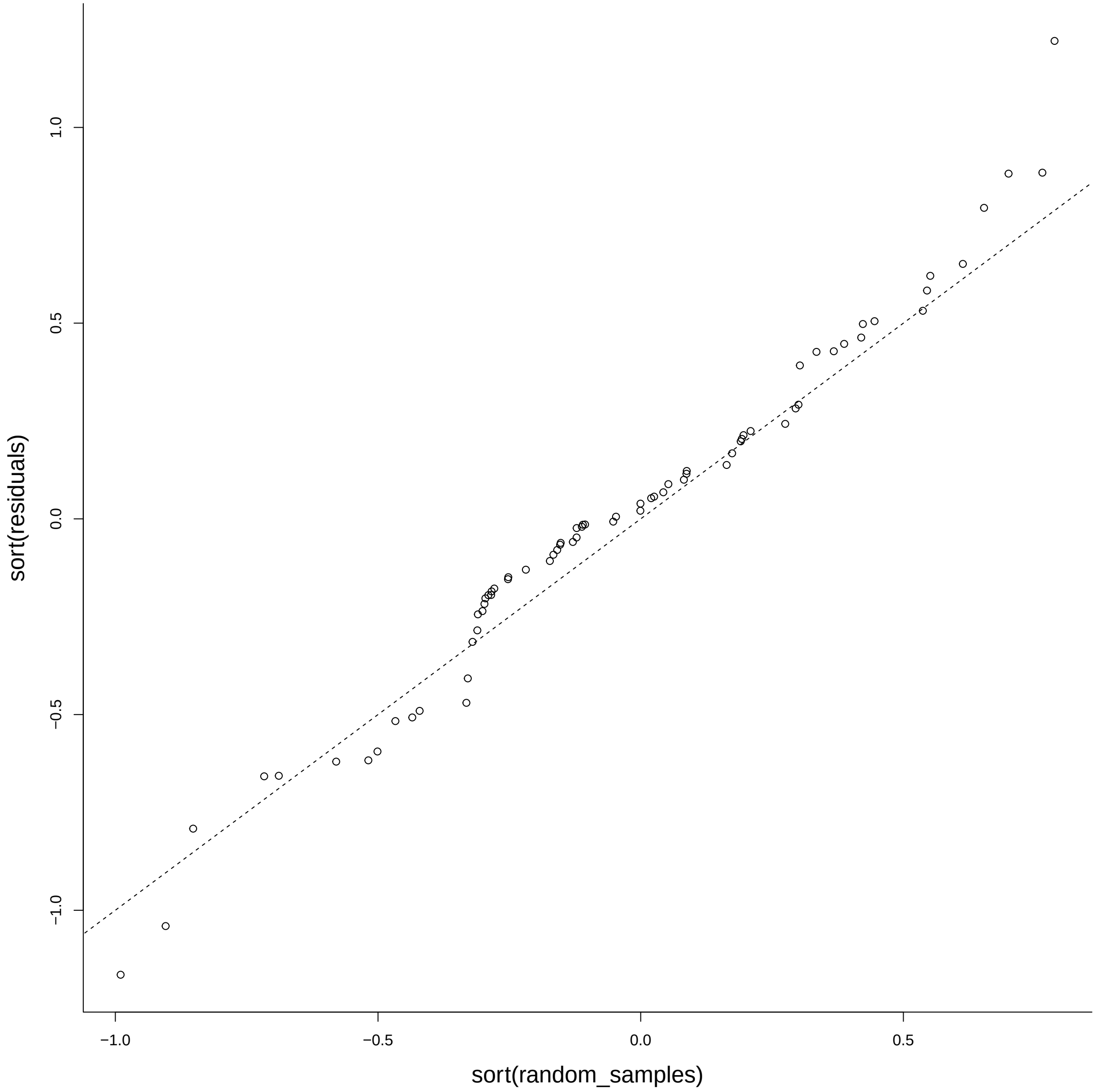

Likelihood
1.0
0.8
0.6
0.4
0.2
0.0
Backcast
Forecast
0.0
0.1
0.2
0.3
0.4
0.5
Regularisation Level (sd2_p)

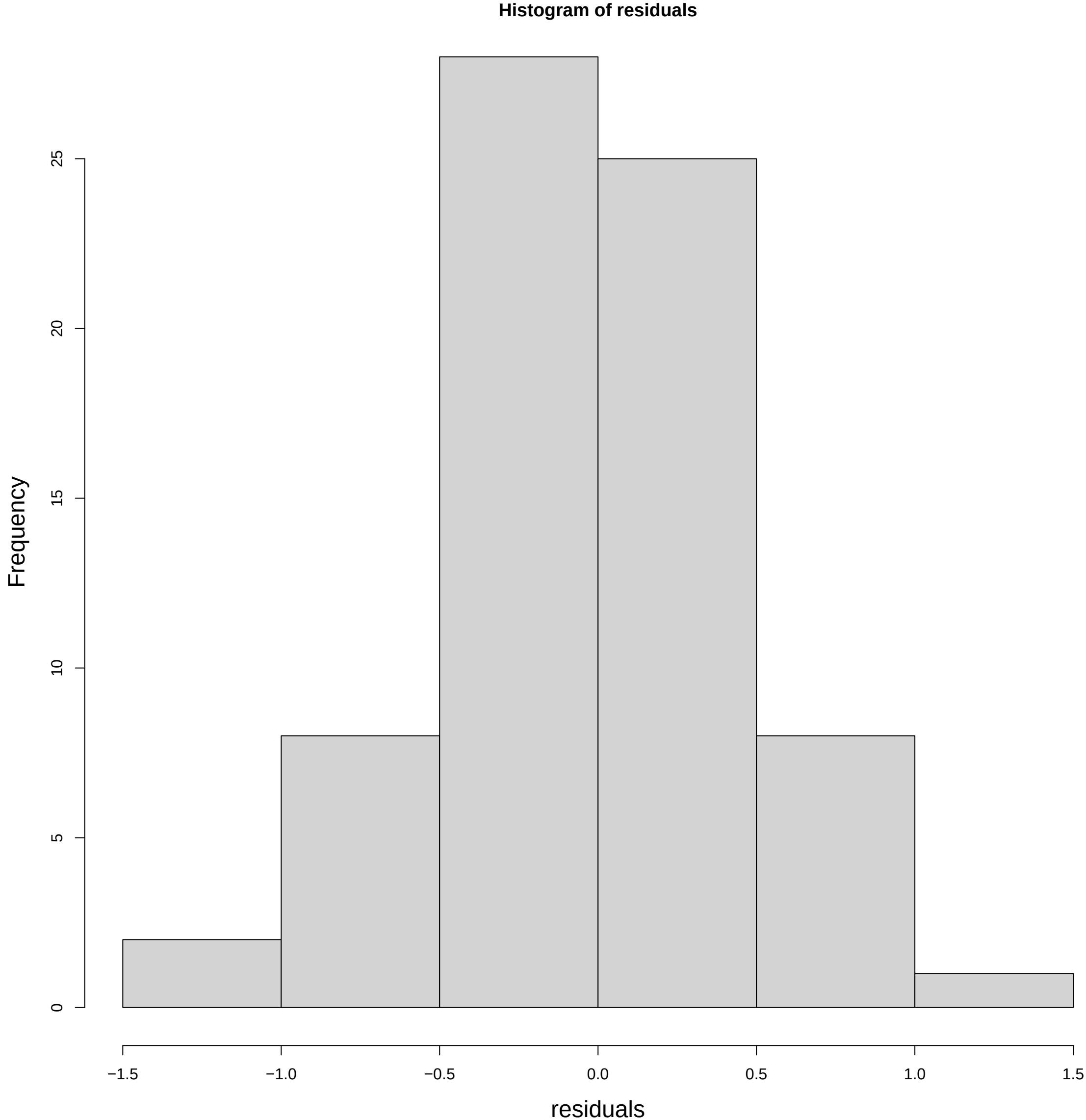

Dynamics
Effects
Time (years)
ΔZ(t)
N:Z
Z:Z
T1:Z
T2:Z
F:Z

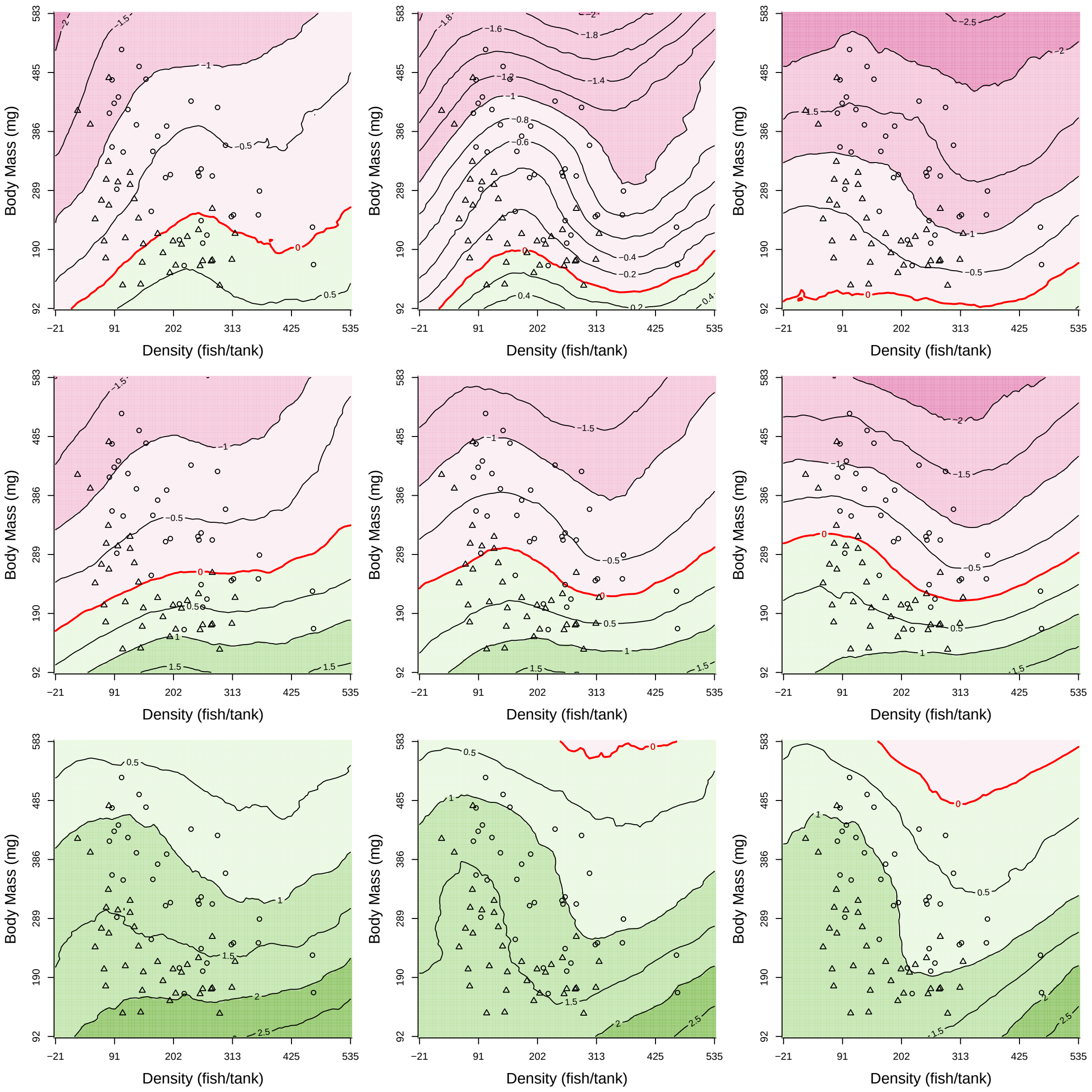

Dynamics
Effects
Time (years)
ΔZ(t)
N:Z
Z:Z
T1:Z
T2:Z
F:Z

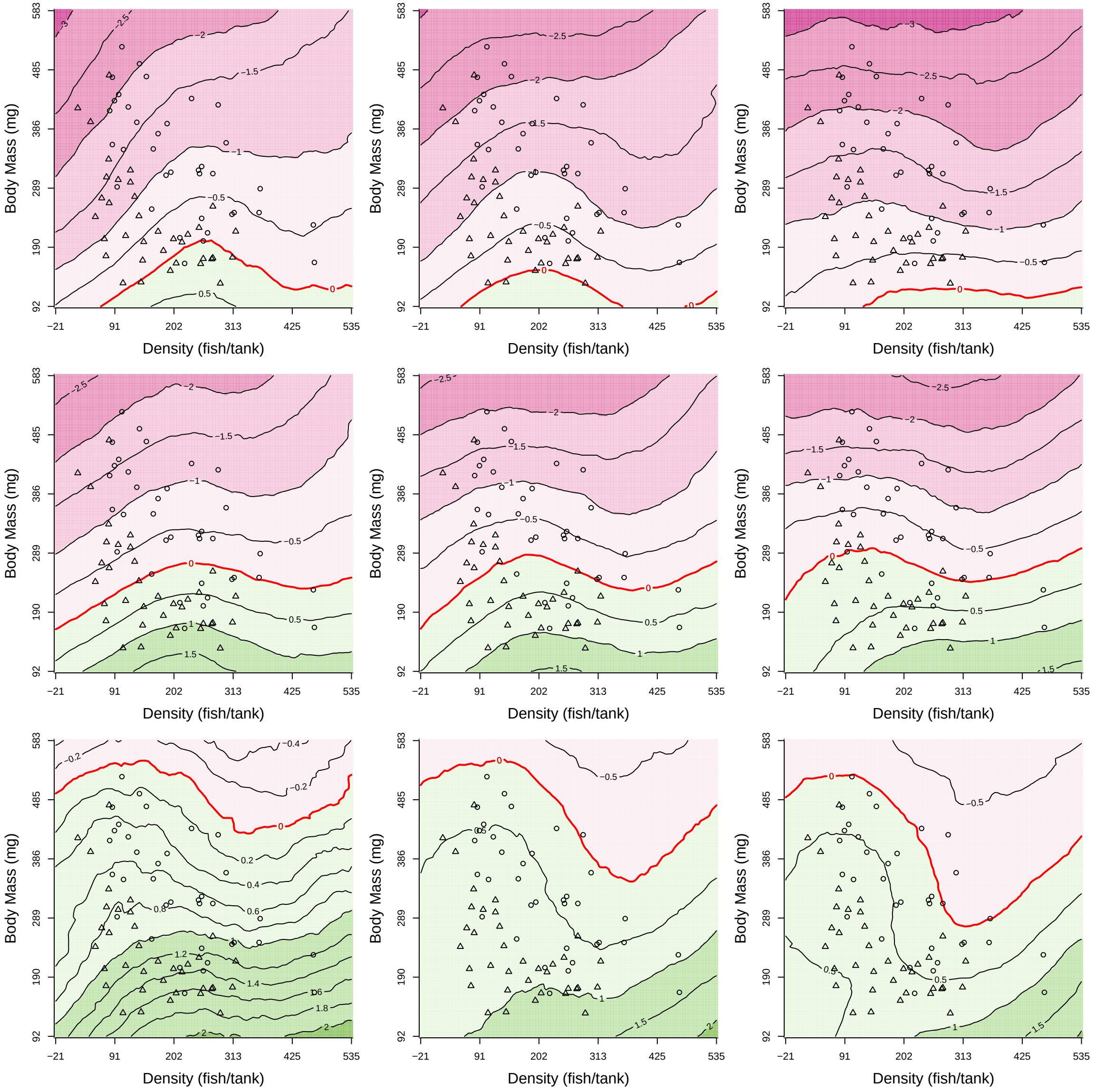

Temperature = −1
Temperature = 1.5
Phenotypic Change
Not Harvested
Harvested
Log Density (S.U.)
Log Body Mass (S.U.)
Summer Temperature (S.U.)
Winter Temperature

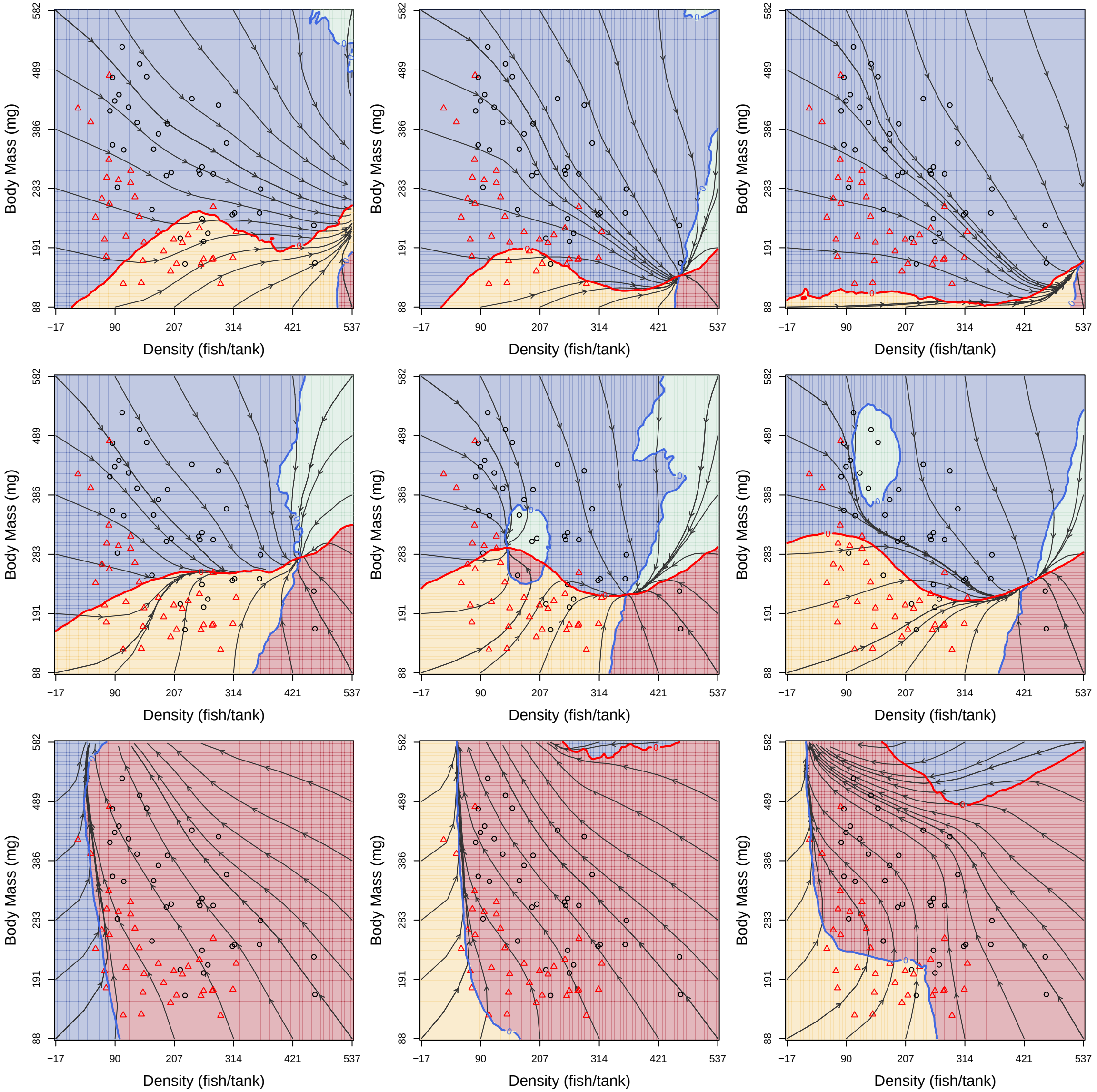

Phenotypic Change
Density (fish/tank)
Not Harvested
Harvested
Body Mass (mg)
Summer Temperature (°C)
Winter Temperature (°C)

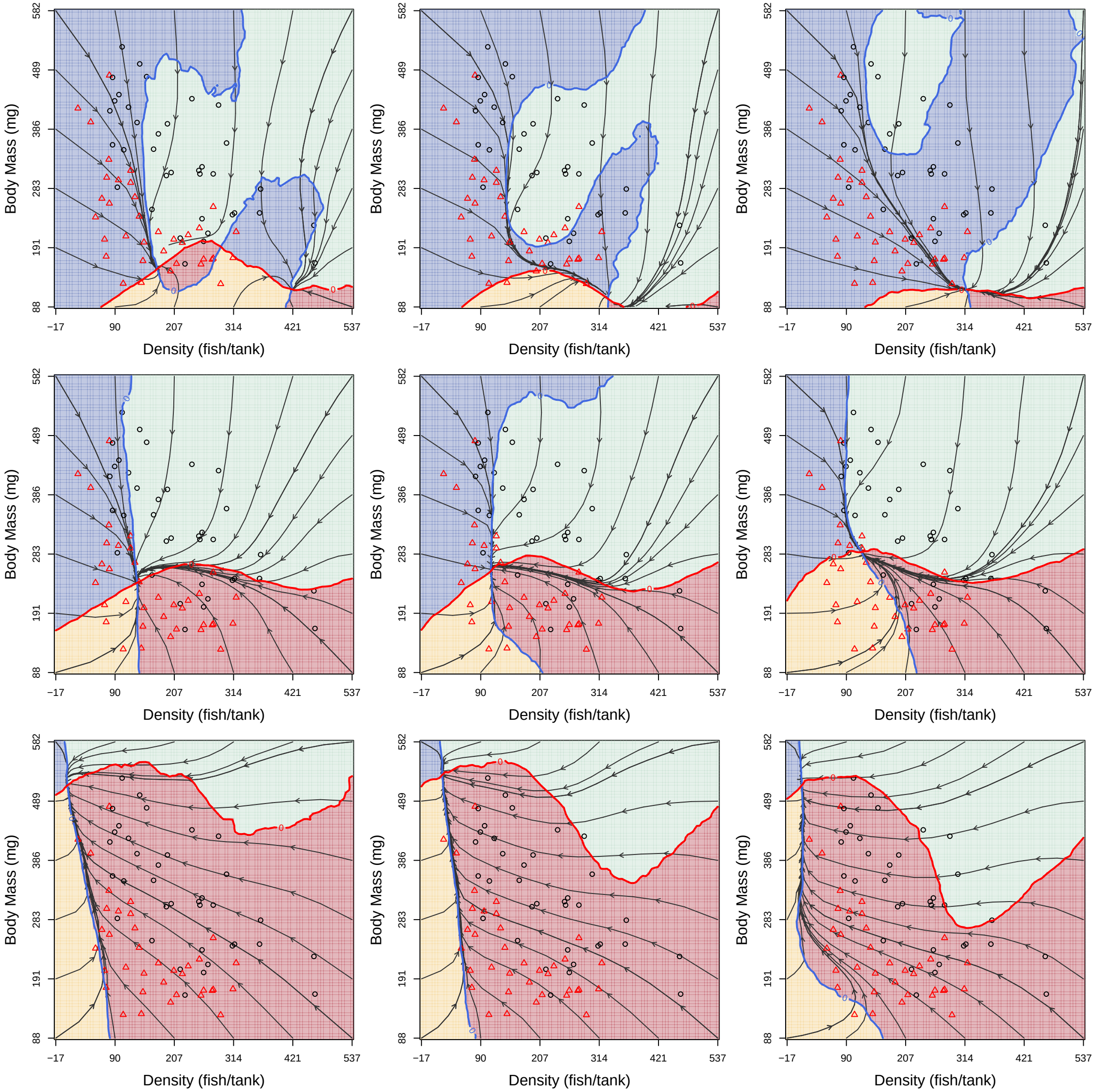

sort(residuals)
1.0
0.5
0.0
−0.5
−1.0
−1.0
−0.5
0.0
0.5
sort(random_samples)

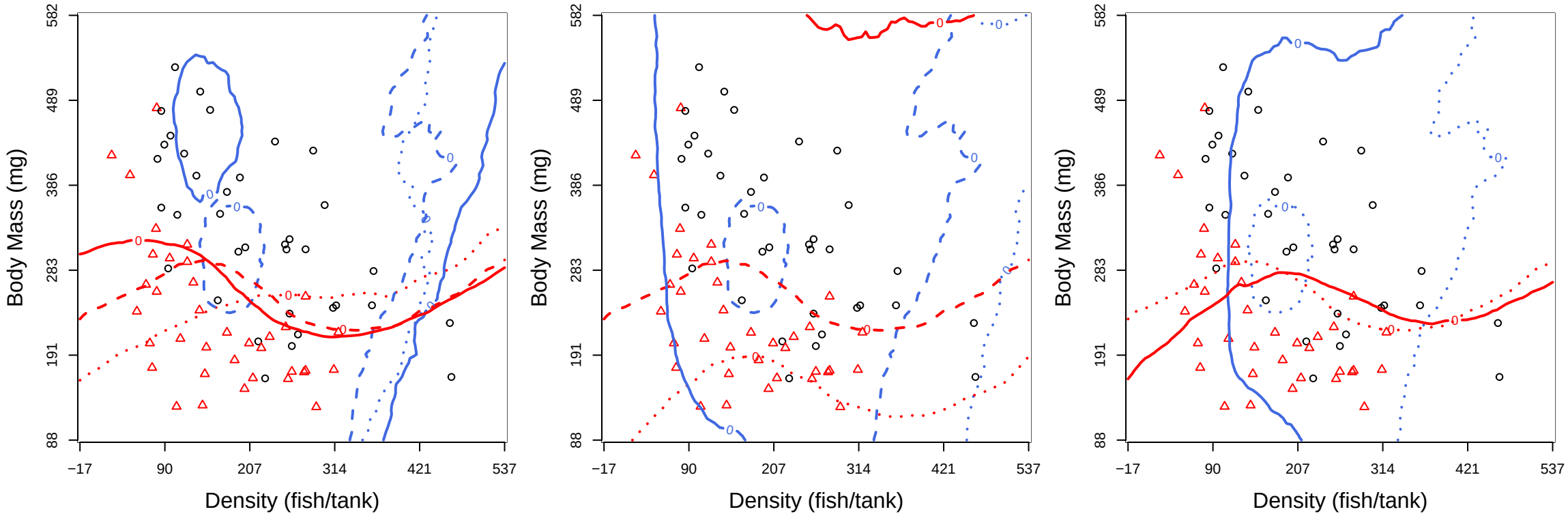

Histogram of residuals
Frequency
0
5
10
15
20
25
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
residuals

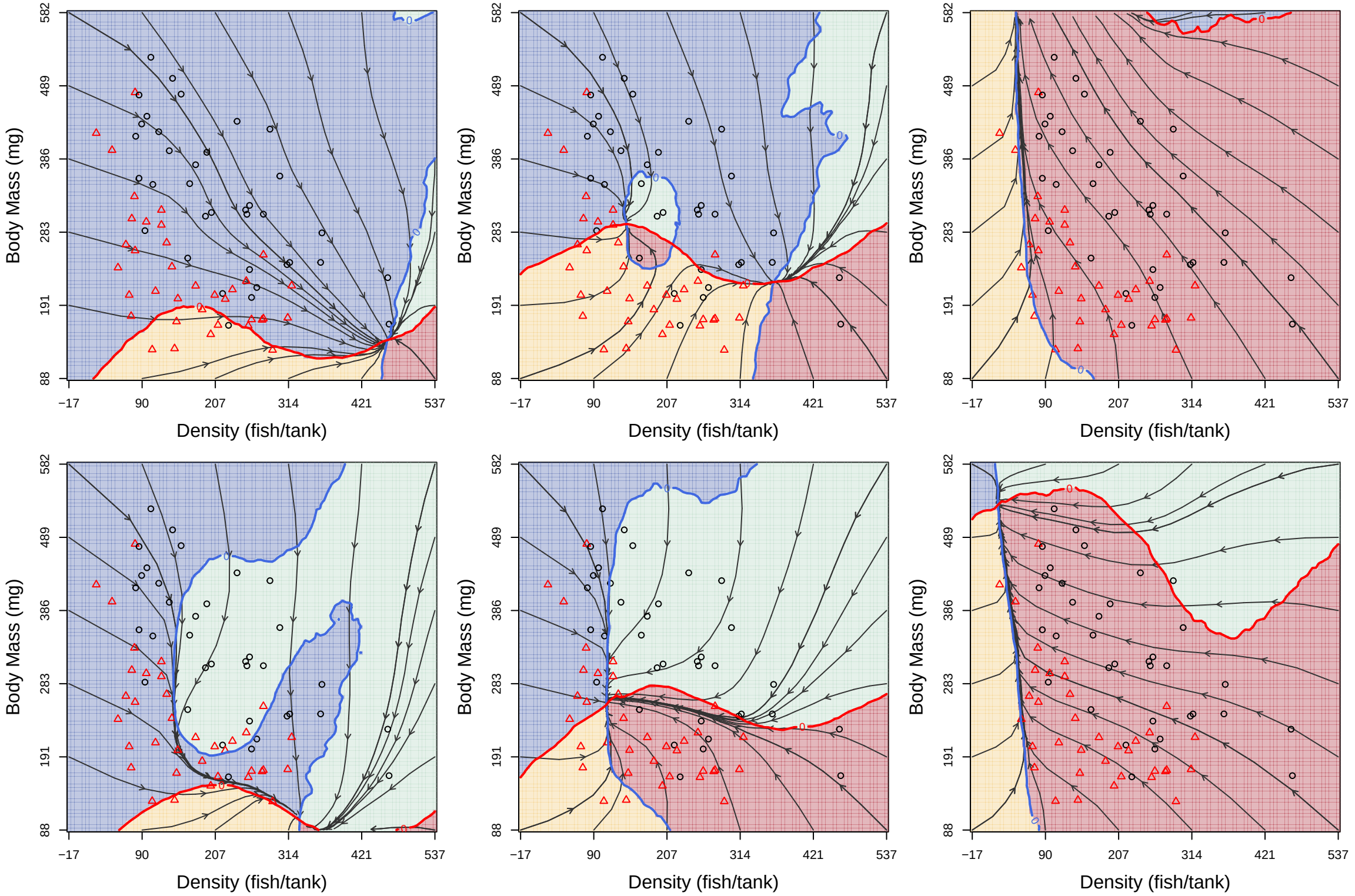

Body Mass (mg)
Density (fish/tank)
583
485
386
289
190
92
−21
91
202
313
425
535
−2
−1.5
−1
−0.5
0
0.5
−1.8
−1.6
−1.4
−1.2
−0.8
−0.6
−0.4
−0.2
0.4
0.2
−2.5
1
1.5
2
2.5


Body Mass (mg)
Density (fish/tank)
583
485
386
289
190
92
−21
91
202
313
425
535


Body Mass (mg)
582
489
386
283
191
88
−17
90
207
314
421
537
Density (fish/tank)
0


Body Mass (mg)
Density (fish/tank)
582
489
386
283
191
88
−17
90
207
314
421
537
0


Body Mass (mg)
582
489
386
283
191
88
−17
90
207
314
421
537
Density (fish/tank)
0


Body Mass (mg)
582
489
386
283
191
88
−17
90
207
314
421
537
Density (fish/tank)
0